

Computer Graphics (CSC214)

Important Topics & Question Frequency Analysis

Based on TU Exam Papers: 2075, 2077, 2078, 2079, 2080, 2081

Frequency Legend:

■ ×1 =

Appeared once

■ ×2 =

Appeared twice

■ ×3 = High priority

■ ×4+ = MUST PREPARE

Unit 1 – Introduction & Graphics Hardware

Raster Scan Display / Vector Display

■ ×2

Years: 2075, 2081

Q: Describe the architecture of raster scan display. Explain vector scan display with suitable diagram.

Raster vs Vector Graphics

■ ×2

Years: 2077, 2079

Q: Differentiate between raster and vector graphics method.

Refresh Rate & Display Devices

■ ×1

Years: 2081

Q: What do you mean by the refresh rate of a display device?

Unit 2 – Scan Conversion Algorithms

Bresenham's Line Drawing Algorithm ★ MOST REPEATED

■ ×5

Years: 2075, 2077, 2079, 2080, 2081

Q: Derive / Trace points using Bresenham's line algorithm. (2081: (6,12)→(10,5); 2080: (1,3)→(6,7); 2079: (6,9)→(2,3); 2075: (1,1)→(8,5); 2077: DDA with (2,2)→(9,2))

DDA Line Algorithm

■ ×2

Years: 2077, 2080

Q: List major differences between DDA and Bresenham's algorithm.

Mid-Point Circle Algorithm

■ ×5

Years: 2075, 2078, 2079, 2080, 2081

Q: Plot first octant / trace mid-point circle algorithm. (2081: center (-2,-2) r=5; 2078: origin r=10; 2080: Q5 algorithm steps)

Mid-Point Ellipse Algorithm

■ ×2

Years: 2075, 2080

Q: Plot ellipse centered (0,0) with rx=8, ry=6; Write algorithm steps.

Boundary Fill & Flood Fill Algorithm

■ ×2

Years: 2075, 2080

Q: Differentiate between boundary and flood fill algorithm.

Scan Line Polygon Fill

■ ×2

Years: 2075, 2078

Q: Describe scan line polygon filling algorithm.

Unit 3 – 2D Geometric Transformations & Clipping

2D Transformations (Translation, Rotation, Scaling, Reflection)

■ ×5

Years: 2075, 2077, 2079, 2080, 2081

Q: Apply transformations with homogeneous matrix. (2081: rotate 90° + translate; 2080: reflect about $X=7$; 2079: reflect about $Y=7$; 2078: magnify $2\times$; 2077: translate by 2,3)

Window to Viewport Transformation

■ ×2

Years: 2075, 2080

Q: Define window, viewport and viewing transformation. Illustrate with example.

Cohen-Sutherland Line Clipping

■ ×2

Years: 2075, 2079

Q: Write algorithm for Cohen-Sutherland. Find region codes and clip given lines.

Liang-Barsky Line Clipping

■ ×2

Years: 2077, 2080

Q: Why is Liang-Barsky more efficient than Cohen-Sutherland? Clip line XY.

Sutherland-Hodgeman Polygon Clipping

■ ×2

Years: 2079, 2081

Q: Clip polygon using Sutherland-Hodgeman algorithm. Drawback + example (2081).

Composite Transformation / Reflection about $y=mx+c$

■ ×2

Years: 2075, 2079

Q: Find composite transformation matrix for reflection about a line $y=mx+c$.

Unit 4 – 3D Geometric Transformations & Projections

Parallel vs Perspective Projection ★

■ ×4

Years: 2077, 2078, 2080, 2081

Q: Define/differentiate orthographic, parallel and perspective projections with diagrams.

3D Translation and Rotation

■ ×2

Years: 2075, 2078

Q: Derive the relation for 3D translation and rotation. Matrix with rotation about X-axis.

Unit 5 – 3D Object Representation & Curves

Bezier Curve & B-Spline Curve ★

■ x5

Years: 2075, 2077, 2078, 2079, 2081

Q: Construct/explain Bezier curve with control points. Properties of Bezier. B-Spline curve and its properties. (2081 control pts: A(1,1) B(2,3) C(4,3) D(6,4))

Polygon Table & Polygon Mesh

■ x4

Years: 2077, 2078, 2079, 2080

Q: How polygon table is used in representing polygons? Explain polygon surface representation using polygon table and meshes.

Hermite Curve

■ x2

Years: 2075, 2077

Q: Explain Hermite curve. Cubic spline interpolation.

Wireframe Representation & Blobby Objects

■ x3

Years: 2075, 2077, 2079

Q: What is the purpose of wireframe representation? Define blobby objects.

Parametric Curves / Spline Representation

■ x3

Years: 2075, 2077, 2079

Q: Explain about parametric curve. Describe spline representation.

Quadric Surface

■ x1

Years: 2078

Q: What is quadric surface? Describe sphere and ellipsoid.

Unit 6 – Solid Modeling

BSP Tree ★

■ x6

Years: 2075, 2077, 2078, 2079, 2080, 2081

Q: Represent polygon using BSP tree. How BSP recursively subdivides space into convex sets. Limitations. (Appeared every year!)

Octree Representation

■ x1

Years: 2075

Q: Explain octree and boundary representations for solid modeling.

Sweep / Boundary Representation

■ x2

Years: 2075, 2081

Q: Explain sweep representation. Boundary and space partitioning.

Unit 7 – Visible Surface Detection

Z-Buffer / Depth Buffer Algorithm ★

■ x4

Years: 2075, 2078, 2079, 2080

Q: Describe Z-buffer hidden surface removal. Limitation of Z-buffer. Compare Scan-line method vs Depth Buffer.

Painter's Algorithm (Depth Sorting)

■ x2

Years: 2075, 2077

Q: Describe Painter's algorithm. Compare Painter's vs A-Buffer algorithm.

BSP Tree Method for Visible Surface

■ x3

Years: 2077, 2078, 2079

Q: How BSP tree is used in visible surface detection?

Ray Tracing

■ x1

Years: 2077

Q: Role of ray tracing in visible surface detection.

Object Space vs Image Space Methods

■ x1

Years: 2079

Q: Differentiate between object space and image space methods.

Scan-Line Algorithm

■ x2

Years: 2077, 2080

Q: Compare Scanline Method with Depth Buffer Method.

Unit 8 – Illumination & Rendering

Phong Shading ★

■ x3

Years: 2079, 2080, 2081

Q: Define rendering. Explain Phong Shading with advantages and disadvantages. Fast Phong Shading.

Gouraud Shading vs Phong Shading

■ x2

Years: 2078, 2079

Q: Distinguish between Gouraud shading and Phong shading model.

Basic Illumination Models (Ambient, Diffuse, Specular)

■ x2

Years: 2077, 2081

Q: Describe any two basic illumination models. Describe Phong model.

Intensity Attenuation

■ x2

Years: 2078, 2081

Q: Define intensity attenuation. Write short note.

Diffuse vs Specular Reflection

■ x1

Years: 2078

Q: Compare between diffuse reflection and specular reflection.

Shadows in Computer Graphics

■ x1

Years: 2075

Q: How can we detect shadows in computer graphics?

Unit 9 – Virtual Reality

Virtual Reality vs Augmented Reality ★

■ x3

Years: 2078, 2080, 2081

Q: Differentiate between virtual reality and augmented reality with examples.

VR Components, Types & Applications

■ x3

Years: 2075, 2077, 2080

Q: Explain components of VR system. Applications of VR. Navigation and manipulation interfaces.

Unit 10 – OpenGL

Drawing in OpenGL (Pixel, Polygon, Line) ★

■ x4

Years: 2075, 2077, 2078, 2081

Q: Give basic commands to draw pixel and polygon in OpenGL. Write procedure to draw a line. Demonstrate polygon creation using OpenGL.

OpenGL Functions & Color Commands

■ x2

Years: 2075, 2081

Q: Explain any two functions in OpenGL. Mention any two color commands.

■ Top Priority Summary (Sorted by Frequency)

Topic	Unit	Freq	Years
BSP Tree	Solid Modeling	■ x6	2075, 2077, 2078, 2079, 2080, 2081
Bresenham's Line Drawing Algorithm	Scan Conversion Algorithms	■ x5	2075, 2077, 2079, 2080, 2081
Mid-Point Circle Algorithm	Scan Conversion Algorithms	■ x5	2075, 2078, 2079, 2080, 2081
2D Transformations (Translation, Rotation, Scaling, Reflection)	2D Geometric Transformations & Clipping	■ x5	2075, 2077, 2079, 2080, 2081
Bezier Curve & B-Spline Curve	3D Object Representation & Curves	■ x5	2075, 2077, 2078, 2079, 2081
Parallel vs Perspective Projection	3D Geometric Transformations & Projections	■ x4	2077, 2078, 2080, 2081
Polygon Table & Polygon Mesh	3D Object Representation & Curves	■ x4	2077, 2078, 2079, 2080
Z-Buffer / Depth Buffer Algorithm	Visible Surface Detection	■ x4	2075, 2078, 2079, 2080
Drawing in OpenGL (Pixel, Polygon, Line)	OpenGL	■ x4	2075, 2077, 2078, 2081
Wireframe Representation & Blobby Objects	3D Object Representation & Curves	■ x3	2075, 2077, 2079
Parametric Curves / Spline Representation	3D Object Representation & Curves	■ x3	2075, 2077, 2079

BSP Tree Method for Visible Surface	Visible Surface Detection	■ ×3	2077, 2078, 2079
Phong Shading	Illumination & Rendering	■ ×3	2079, 2080, 2081
Virtual Reality vs Augmented Reality	Virtual Reality	■ ×3	2078, 2080, 2081
VR Components, Types & Applications	Virtual Reality	■ ×3	2075, 2077, 2080
Raster Scan Display / Vector Display	Introduction & Graphics Hardware	■ ×2	2075, 2081
Raster vs Vector Graphics	Introduction & Graphics Hardware	■ ×2	2077, 2079
DDA Line Algorithm	Scan Conversion Algorithms	■ ×2	2077, 2080

★ Topics marked with ★ are extremely high priority — prepare thoroughly with numerical examples.